

KUNMING WUJIABA, China

WMO: 567780

Lat: 24.992N

Lon: 102.744E

Elev: 1895

StdP: 80.54

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	0.1	1.9	-6.8	2.7	16.4	-4.9	3.1	15.7	10.6	15.3	9.6	14.2	3.1	40	0.406

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	7.9	28.1	16.4	27.1	16.4	26.1	16.5	19.7	24.4	19.4	23.8	19.0	23.1	5.0	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.5	16.9	21.2	18.1	16.5	20.9	17.8	16.2	20.7	66.1	24.8	64.4	23.9	63.1	23.3	23.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.0	8.6	7.5	DB	-2.3	30.1	2.1	1.0	-3.8	30.8	-5.0	31.4	-6.1	32.0	-7.6	32.8	(4)
(5)			WB	-3.2	21.1	2.0	0.9	-4.7	21.7	-5.8	22.2	-6.9	22.7	-8.4	23.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	16.2	9.8	11.9	15.0	18.1	19.7	20.7	20.6	20.4	18.9	16.5	12.9	9.8	(6)	
	DBStd	4.74	2.57	3.21	3.04	2.86	2.85	2.07	1.52	1.45	2.24	2.52	2.47	2.74	(7)	
	HDD10.0	96	31	18	6	1	0	0	0	0	0	1	6	34	(8)	
	HDD18.3	1126	263	181	106	37	21	6	2	2	19	64	162	264	(9)	
	CDD10.0	2369	26	71	161	243	299	322	329	323	268	203	94	29	(10)	
	CDD18.3	358	0	0	3	29	62	78	73	67	37	8	0	0	(11)	
	CDH23.3	1382	0	3	42	242	377	272	173	171	84	17	1	0	(12)	
(13)	CDH26.7	136	0	0	1	23	59	32	9	9	3	0	0	0	(13)	
(14)	Wind	WSAvg	3.2	3.4	3.9	4.1	4.0	3.6	3.1	2.6	2.4	2.5	2.7	2.8	3.0	(14)
(15) Precipitation	PrecAvg	980	14	12	18	27	83	176	210	199	120	84	36	14	(15)	
	PrecMax	1387	51	48	77	122	220	475	368	410	283	293	148	69	(16)	
	PrecMin	567	0	0	0	0	8	36	73	57	28	9	0	0	(17)	
	PrecStd	193	12	13	18	24	53	81	82	77	56	55	32	15	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.8	23.7	26.0	28.8	29.9	29.1	27.8	27.8	27.1	25.0	22.8	20.2	(19)	
		MCWB	8.6	9.4	11.3	13.4	16.0	18.2	19.1	18.9	18.1	16.2	12.9	10.0	(20)	
	2%	DB	19.0	21.9	24.4	27.1	28.2	27.5	26.2	26.2	25.3	23.7	20.9	18.7	(21)	
		MCWB	8.5	9.4	11.0	13.2	15.5	18.1	18.9	18.5	17.7	15.8	12.2	9.4	(22)	
	5%	DB	17.8	20.3	23.1	26.0	27.0	26.1	25.2	25.2	24.1	22.2	19.5	17.2	(23)	
		MCWB	8.2	9.0	10.7	12.9	15.6	18.1	18.7	18.2	17.2	15.3	11.8	9.2	(24)	
	10%	DB	16.2	19.1	22.0	24.8	25.8	24.9	24.1	24.0	23.0	21.1	18.1	15.8	(25)	
		MCWB	7.7	8.7	10.5	12.8	15.5	18.1	18.4	18.1	16.9	15.2	11.5	8.9	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.9	11.6	13.3	16.0	19.1	20.1	20.5	20.2	19.6	17.9	15.5	12.5	(27)	
		MCDB	16.2	18.8	20.5	23.6	24.6	25.1	24.6	25.3	24.5	22.4	18.7	16.3	(28)	
	2%	WB	9.9	10.7	12.5	15.1	18.3	19.5	19.8	19.6	18.8	17.2	14.4	11.1	(29)	
		MCDB	15.6	18.1	20.0	22.6	23.2	24.3	24.2	24.2	23.1	21.1	17.7	15.5	(30)	
	5%	WB	9.2	9.9	11.9	14.5	17.7	19.1	19.4	19.1	18.3	16.6	13.5	10.2	(31)	
		MCDB	15.1	17.7	19.6	21.9	22.5	23.6	23.5	23.3	22.2	20.3	17.0	14.9	(32)	
	10%	WB	8.5	9.3	11.3	13.8	17.1	18.7	18.9	18.6	17.8	16.1	12.5	9.5	(33)	
		MCDB	14.3	17.0	19.0	21.3	21.9	22.8	22.7	22.4	21.4	19.4	16.3	14.1	(34)	
(35) Mean Daily Temperature Range	MDBR		12.5	12.9	12.6	12.0	10.0	7.9	7.2	7.9	8.0	8.5	10.5	11.2	(35)	
	5% DB	MCDBR	14.9	15.1	14.5	13.6	12.4	10.0	9.1	9.7	9.7	10.4	12.4	13.7	(36)	
		MCWBR	6.1	5.7	5.1	4.3	3.8	3.5	3.6	3.7	3.8	4.2	5.1	5.8	(37)	
	5% WB	MCDBR	13.1	13.5	12.6	11.9	9.2	8.0	7.6	8.3	8.3	8.3	9.9	11.5	(38)	
		MCWBR	6.1	5.7	5.0	4.5	3.7	3.3	3.4	3.5	3.6	3.8	4.9	5.5	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.336	0.350	0.500	0.491	0.424	0.408	0.422	0.460	0.439	0.417	0.370	0.361	(40)	
	taud		2.365	2.334	1.871	1.967	2.268	2.354	2.300	2.164	2.206	2.229	2.300	2.285	(41)	
	Ebn at Noon		918	930	802	820	872	880	868	836	848	847	871	871	(42)	
	Edh at Noon		108	120	201	187	138	126	133	151	142	132	115	113	(43)	
(44) All-Sky Solar Radiation	RadAvg		3.84	4.82	5.18	5.56	5.27	4.49	4.30	4.34	3.85	3.43	3.76	3.31	(44)	
	RadStd		0.38	0.59	0.46	0.50	0.61	0.48	0.31	0.31	0.32	0.41	0.42	0.39	(45)	

Historical Trends

		DBAvg		Heating		Cooling			Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	-0.97	+0.64	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (8 neighbors)	+0.42	N/A	N/A	+0.77	N/A	-0.22	N/A	N/A	+106	+60	(47)

Nomenclature: See separate page